

The two Rayleigh–Ritz discriminant degrees of the standard rational normal curve

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Abstract

For the standard unweighted rational normal curve of degree $d \geq 2$, we prove that the reduced isotropic part of its Rayleigh–Ritz discriminant has degree $2d$ and its reduced nonisotropic part has degree $6(d - 1)$. The argument identifies the critical correspondence with a binary Wronskian, establishes the irreducibility of the nonisotropic image, and proves that every component occurs with multiplicity one in the binary discriminant. It includes the smallest degree, the points at infinity, and the kernel of the map from symmetric matrices to binary numerators.

1. Statement

Let

$$C_d = \{[a^d : a^{d-1}b : \dots : b^d] : [a : b] \in \mathbb{P}^1\} \subset \mathbb{P}^d, \quad d \geq 2.$$

All spaces and algebraic sets in this note are over $\mathbb{C}$. Transposes do not conjugate. For a symmetric matrix H of order $d + 1$, consider

$$\rho_H(z) = \frac{z^\top H z}{z^\top z}$$

on the part of C_d with nonzero denominator. Let $\mathcal{R}$ be the Zariski closure of its critical-point correspondence in $C_d \times \mathbb{P}(\text{Sym}_{d+1}(\mathbb{C}))$. For homogeneous generators g_i of the ideal of $\mathcal{R}$, put

$$\mathcal{B} = \{(z, H) \in \mathcal{R} : \text{rank}(\partial g_i / \partial z_j) \leq d - 1\}.$$

With $Q = \{z^\top z = 0\}$ and projection π to matrix space, define

$$\Sigma_{\text{iso}} = \pi(\mathcal{B} \cap Q), \quad \Sigma_{\text{noniso}} = \overline{\pi(\mathcal{B} \setminus Q)}.$$

Degrees refer to the reduced projective algebraic sets, including all their components.

Theorem 1. Both sets are hypersurfaces. The isotropic set is a union of $2d$ distinct hyperplanes, the nonisotropic set is irreducible, and

$$\deg \Sigma_{\text{iso}} = 2d, \quad \deg \Sigma_{\text{noniso}} = 6(d - 1).$$

This is the precise target of repository problem TR-26 and Conjecture 3.14 of [1]. Proposition 3.10 of [1] states the total discriminant degree $8d - 6$. Our proof derives both the total degree and the reduced component degrees, including the required multiplicity assertions.

2. Binary forms and the exact correspondence

Put $m = 2d$ and $V = H^0(\mathbb{P}^1, \mathcal{O}(m))$. The denominator pulled back to the parameter line is

$$D(a, b) = \sum_{i=0}^d a^{m-2i} b^{2i}.$$

The linear map

$$T : \text{Sym}_{d+1}(\mathbb{C}) \longrightarrow V, \quad H \longmapsto N_H = z^\top H z$$

is surjective. In fact, the coefficient of $a^{m-k}b^k$ depends only on the entries on the antidiagonal $i + j = k$, indexing from zero. Each $0 \leq k \leq 2d$ occurs, and these antidiagonals are disjoint. Diagonal entries have coefficient one and off-diagonal entries have coefficient two, both nonzero over $\mathbb{C}$.

In the chart $a = 1$,

$$D(t) = 1 + t^2 + \dots + t^m = \frac{t^{m+2} - 1}{t^2 - 1}.$$

Its roots are precisely the $(m+2)$ -th roots of unity other than $1, -1$. Thus there are m distinct roots, all finite and nonzero. Neither coordinate endpoint of $\mathbb{P}^1$ is a zero. Write $Z = \{D = 0\}$.

For $N \in V$, define the binary form

$$F_N(a, b) = \frac{N_b D_a - N_a D_b}{m}.$$

It has homogeneous degree $2m - 2$ and depends linearly on N . Euler's identity gives

$$F_N(1, t) = N'(t)D(t) - N(t)D'(t). \quad (1)$$

On $D \neq 0$, its zero set is therefore the critical set of N/D .

Lemma 2. Over either numerator projective space or matrix projective space, the closure of the critical-point correspondence is exactly the equation $F_N(p) = 0$. It is a smooth irreducible projective hyperplane bundle over $\mathbb{P}^1$.

Proof. For every $p \in \mathbb{P}^1$, the functional $N \mapsto F_N(p)$ is nonzero. When $D(p) \neq 0$, freely prescribing $N(p)$ and $N'(p)$ in a local coordinate proves this. At a root of D , equation (1) gives $F_N(p) = -D'(p)N(p)$, with $D'(p) \neq 0$.

The resulting surjective bundle map from the trivial bundle with fiber V to $\mathcal{O}(2m-2)$ has a locally free kernel. Its projectivization is the stated smooth irreducible incidence. Its part over $\mathbb{P}^1 \setminus Z$ is dense, so it is exactly the closure of the critical correspondence. Surjectivity of T proves the same facts directly in matrix space. Finally, the rational normal parametrization is an isomorphism from $\mathbb{P}^1$ onto C_d . $\square$

Lemma 3. Under that parametrization, the set $\mathcal{B}$ in Theorem 1 is the set

$$F_N(p) = 0, \quad \partial_t F_N(p) = 0 \quad (2)$$

in any local coordinate and local trivialization at p .

Proof. The smooth curve has $d - 1$ locally independent normal equations in an affine chart of $\mathbb{P}^d$. By Lemma 2, adjoining the equation $F_N(t) = 0$ gives the local ideal of the correspondence. Holding

the matrix parameters fixed, its Jacobian has rank d if $\partial_t F_N \neq 0$, and rank $d - 1$ otherwise. This calculation is independent of the choice of generators: on their common zero set, the differentials of any ideal-generating set span the same conormal subspace.

In homogeneous coordinates the radial vector lies in the Jacobian kernel by Euler's identity. Dehomogenization removes that direction without changing the rank, giving exactly the threshold used in Theorem 1. A change of local trivialization multiplies F_N by a unit, so on $F_N = 0$ the derivative condition is unchanged. This proves (2), including at infinity and when F_N is identically zero. $\square$

We first compute the two images in $\mathbb{P}(V) = \mathbb{P}^m$ and return to matrix space in Section 7.

3. The isotropic hyperplanes

Fix $r \in Z$ and use a local coordinate s with $r = 0$. Write

$$D = d_1 s + d_2 s^2 + \cdots, \quad d_1 \neq 0, \quad N = n_0 + n_1 s + n_2 s^2 + \cdots.$$

At r , the equation $F_N(r) = 0$ is equivalent to $n_0 = 0$. Under this condition,

$$F_N = (n_2 d_1 - n_1 d_2) s^2 + O(s^3). \quad (3)$$

Thus every pair in the correspondence above an isotropic point is ramified. Its image is precisely the hyperplane

$$L_r = \{[N] \in \mathbb{P}(V) : N(r) = 0\}.$$

Evaluation at distinct points gives nonproportional linear functionals on the full space of binary forms. Hence these m hyperplanes are distinct, and

$$\Sigma_{\text{iso}} = \bigcup_{r \in Z} L_r, \quad \deg \Sigma_{\text{iso}} = m. \quad (4)$$

This is a set equality for every numerator, not merely a generic assertion.

4. The nonisotropic image

Let $U = \mathbb{P}^1 \setminus Z$. With $f = N/D$, define

$$Y = \{(p, [N]) \in U \times \mathbb{P}(V) : f'(p) = f''(p) = 0\}. \quad (5)$$

Since $F_N = D^2 f'$, these are exactly the conditions (2) on U .

We use the following elementary interpolation fact. For distinct points with prescribed jets of total length ℓ , the restriction map from $H^0(\mathbb{P}^1, \mathcal{O}(m))$ is surjective if $m \geq \ell - 1$. Choose a projective coordinate whose infinity avoids the specified points. The assertion is then polynomial Hermite interpolation, or the Chinese remainder theorem modulo the corresponding powers of distinct linear factors. A section of degree at most $\ell - 1$ already realizes every prescription.

The two conditions in (5) are independent linear conditions: the full 2-jet of N is arbitrary, and division by nonzero D is invertible on jets. Consequently Y is a projective bundle with fiber $\mathbb{P}^{m-2}$ over U , and is irreducible of dimension $m - 1$.

The only N with $F_N \equiv 0$ are $N = cD$, because a rational function with identically zero derivative is constant in characteristic zero. Every other N has a finite fiber in (5), as a nonzero binary form has finitely many roots. The incidence is not supported over $N = cD$: for example, a numerator

vanishing to order exactly three at a point of U lies in (5) and is not proportional to D . The projection is thus generically finite onto its image. Its closure S is an irreducible hypersurface of dimension $m - 1$ in $\mathbb{P}^m$.

By Lemma 3, S is exactly the reduced nonisotropic image in numerator space.

For each $r \in Z$, one has $S \neq L_r$. Choose $p \in U$. Length-four interpolation provides an N with vanishing value and first two derivatives at p , but with $N(r) \neq 0$. Its image belongs to $S \setminus L_r$. Since both are irreducible hypersurfaces, they are distinct. In particular, no isotropic hyperplane is a component of the nonisotropic closure.

5. Generic multiple-root patterns

Lemma 4. A general point of S has exactly one multiple root of F_N ; it is exactly double and nonisotropic. All other roots are simple.

Proof for $m \geq 6$. In (5), the additional condition $f'''(p) = 0$ is independent of the other two by length-four interpolation. This incidence has dimension $m - 2$.

For distinct $p, q \in U$, the four conditions

$$f'(p) = f''(p) = f'(q) = f''(q) = 0$$

are independent. Prescribe both entire 2-jets, using length-six interpolation, valid for $m \geq 5$, and divide by the nonvanishing denominator jets. The ordered-pair incidence has projective fibers of dimension $m - 4$ over a two-dimensional base, so its total dimension is $m - 2$. The closures of both exceptional images therefore have dimension at most $m - 2$, strictly below $\dim S$.

A general point of S avoids these loci and all the distinct hypersurfaces L_r . Its one nonisotropic multiple root is exactly double, it has no other nonisotropic multiple root, and it has no isotropic root. These are statements about the full projective parameter line and include infinity. $\square$

Proof for $m = 4$. Length-six interpolation is unavailable here. For the specified denominator, take

$$D(t) = 1 + t^2 + t^4, \quad N(t) = t^3.$$

Then

$$F_N(t) = t^2(3 + t^2 - t^4). \tag{6}$$

The root at zero is exactly double and nonisotropic. The quartic factor has four distinct nonzero roots, because $3 + u - u^2$ has two distinct nonzero roots, each giving two distinct square roots. The coefficient of t^6 is nonzero, so infinity is not a root. At any zero r of D , one has $F_N(r) = -r^3 D'(r) \neq 0$.

Within the space of degree-six binary forms with a multiple root, those with precisely one exactly double root and all remaining roots simple form an open subset. To justify openness, its complement is the union of the closed locus of forms divisible by a cube of a linear factor and the closed locus of forms divisible by squares of two linear factors. These are images of projective parameter spaces; allowing the two factors to coincide adds only fourth-order roots, already in the first locus. The example (6) makes the pullback of this open set a nonempty open subset of the irreducible S . A general point of S therefore has the asserted property. $\square$

Lemma 5. A general point of L_r , for each $r \in Z$, has an exactly double root at r and all other roots simple.

Proof. Such a point avoids S and all $L_{r'}$ with $r' \neq r$, since these are distinct hypersurfaces. Every multiple root outside Z would put the numerator in S , and any root at another $r' \in Z$ would put it in $L_{r'}$. Equation (3) shows that the root at r is exactly double unless $n_2 d_1 - n_1 d_2 = 0$. This is a proper linear condition inside L_r : one can choose $n_1 = 0, n_2 = 1$ by jet interpolation. $\square$

6. Discriminant multiplicities and degrees

Let $k = 2m - 2$, and form the polynomial

$$\Delta(N) = \text{Disc}_k(F_N).$$

The binary discriminant vanishes exactly on forms with a multiple projective root, including the zero form. Lemma 3 and Sections 3–4 give

$$V(\Delta) = S \cup \bigcup_{r \in Z} L_r. \quad (7)$$

This union is proper, so Δ is not identically zero.

For completeness, the discriminant of a degree- k binary form is homogeneous of degree $2k - 2$ in its coefficients. In a coordinate with nonzero leading coefficient and finite roots, this is the polynomial extending

$$c^{2k-2} \prod_{i < j} (\alpha_i - \alpha_j)^2$$

for $c \prod_i (t - \alpha_i)$. Equivalently it is, up to sign, the resultant with the derivative divided by the leading coefficient. It extends as a polynomial to all binary forms and vanishes precisely on projective multiple-root forms. Scaling every coefficient scales it by the power $2k - 2$, giving the stated degree.

The coefficients of F_N are linear homogeneous functions of the coefficients of N . Thus the nonzero homogeneous polynomial Δ has degree

$$2k - 2 = 4m - 6. \quad (8)$$

This is not yet a reduced degree assertion. We next prove that every component in (7) has multiplicity one.

Lemma 6. At a binary form F with exactly one double root p and all other roots simple, the differential of its discriminant is a nonzero scalar multiple of the functional $\dot{F} \mapsto \dot{F}(p)$, after choosing a local trivialization at p .

Proof. Choose a coordinate making all roots finite, translate p to zero, and restrict to a neighborhood with nonzero leading coefficient. The simple roots depend analytically on coefficients, by the implicit function theorem, and can be factored off. The remaining monic quadratic is $t^2 + ut + v$, with $u = v = 0$ at the original form. The discriminant product formula shows that the full discriminant is a nonvanishing analytic factor times $u^2 - 4v$. Its differential is a nonzero multiple of $\dot{v}$. Evaluating the factored polynomial at zero shows that $\dot{F}(0)$ is a nonzero multiple of $\dot{v}$, proving the assertion. $\square$

At a general point of S , Lemma 4 and Lemma 6 apply. In the numerator family the resulting functional is

$$\dot{F}_N(p) = D(p)\dot{N}'(p) - D'(p)\dot{N}(p),$$

which is nonzero as a linear functional of $\dot{N}$, by first-jet interpolation. At a general point of L_r , Lemma 5 applies, and the functional is

$$\dot{F}_N(r) = -D'(r)\dot{N}(r),$$

again nonzero. These nonzero directions remain valid in projective parameter space: the radial direction gives zero, since $F_N(p) = 0$.

It follows that the pulled-back discriminant has nonzero differential at a general point of every component in (7). Thus each irreducible factor occurs to exponent one. If G is an irreducible reduced defining polynomial of S , then, up to nonzero scalar,

$$\Delta(N) = G(N) \prod_{r \in Z} N(r). \quad (9)$$

Taking degrees in (8)–(9) gives

$$\deg S = (4m - 6) - m = 3m - 6. \quad (10)$$

7. Matrix space and degenerate numerator parameters

Choose a linear splitting of the surjection T from Section 2. Pulling a polynomial back along T simply adjoins independent kernel variables to the polynomial ring in the numerator coefficients. An irreducible polynomial over $\mathbb{C}$ remains irreducible after adjoining variables. The evaluation hyperplanes remain distinct, and all degrees are preserved.

There is a potential issue if one uses only the projective rational map $[H] \dashrightarrow [N_H]$: it is undefined at nonzero elements of $\ker T$. This issue is avoided by working with affine cones and the surjective linear map T .

For nonzero N , the preceding incidences and images pull back to the corresponding matrix incidences, by Lemmas 2–3. Their closures pull back as well: in a linear splitting the affine preimage is a product with the kernel, so the preimage of a dense set is dense in the preimage of its closure. At $T(H) = 0$, the form F_{N_H} is identically zero. Every point of the curve then belongs to the correspondence and satisfies the ramification condition. Since both Z and its complement are nonempty, such matrix parameters belong to both images, as their pulled-back homogeneous equations prescribe. These are cone vertices, not additional components. Parameters with $N_H = cD$, giving constant Rayleigh quotient, are likewise already included in both incidences.

Consequently (4) and (10) transfer to the exact matrix projective space of Theorem 1. Substituting $m = 2d$ yields

$$\deg \Sigma_{\text{iso}} = 2d, \quad \deg \Sigma_{\text{noniso}} = 6(d - 1),$$

with the claimed component statements. This completes the proof of Theorem 1.

8. Scope

The denominator's simple roots are essential. The proof is for the fixed unweighted embedding and bilinear form specified in Theorem 1, and does not silently extend to weighted embeddings with repeated denominator roots. The degree-four numerator example is used only for $d = 2$; the higher orders are proved by interpolation and dimension arguments. No finite computational experiment is used to justify an all-order step.

References

1. V. Borovik, H. Friedman, S. Hoşten, and M. Pfeffer, *Numerical Algebraic Geometry for Energy Computations on Tensor Train Varieties*, arXiv:2512.06939v2, 18 June 2026, §3.2, Propositions 3.6–3.10 and Conjecture 3.14.
2. OpenProblemsInNLA, TR-26: Degrees of the two Rayleigh–Ritz discriminant parts for rational normal curves, canonical problem statement consulted 11 September 2026.